

Ex: If matrices $A = \begin{bmatrix} 9 & 4 & 6 & 7 \\ 7 & 8 & 1 & 4 \\ 4 & 3 & 2 & 6 \\ 5 & 3 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 7 & 6 & 2 & 1 \\ 3 & 9 & 0 & 3 \\ 2 & 5 & 2 & 9 \\ 3 & 2 & 4 & 7 \end{bmatrix}$

Implement Strassen's matrix multiplication on $A \times B$.

Sol:
 $A = \begin{bmatrix} 9 & 4 & 6 & 7 \\ 7 & 8 & 1 & 4 \\ 4 & 3 & 2 & 6 \\ 5 & 3 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 7 & 6 & 2 & 1 \\ 3 & 9 & 0 & 3 \\ 2 & 5 & 2 & 9 \\ 3 & 2 & 4 & 7 \end{bmatrix}$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix}$$

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Steps: Divide the given 4×4 matrices into submatrices of

Size 2×2

$$A = \begin{array}{cc|cc} & A_{11} & & A_{12} & \\ \hline 9 & 4 & 6 & 7 \\ 7 & 8 & 1 & 4 \\ \hline 4 & 3 & 2 & 6 \\ 5 & 3 & 0 & 2 \\ \hline & A_{21} & & A_{22} \end{array}$$

$$A_{11} = \begin{bmatrix} 9 & 4 \\ 7 & 8 \end{bmatrix} \quad A_{12} = \begin{bmatrix} 6 & 7 \\ 1 & 4 \end{bmatrix} \quad A_{21} = \begin{bmatrix} 4 & 3 \\ 5 & 3 \end{bmatrix} \quad A_{22} = \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix}$$

$$B = \begin{array}{cc|cc} & B_{11} & & B_{12} & \\ \hline 7 & 6 & 2 & 1 \\ 3 & 9 & 0 & 3 \\ \hline 2 & 5 & 2 & 9 \\ 3 & 2 & 4 & 7 \\ \hline & B_{21} & & B_{22} \end{array}$$

$$B_{11} = \begin{bmatrix} 7 & 6 \\ 3 & 9 \end{bmatrix} \quad B_{12} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \quad B_{21} = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \quad B_{22} = \begin{bmatrix} 2 & 9 \\ 4 & 7 \end{bmatrix}$$

The submatrices of size 2×2 can be computed as follows

(i) calculate $C_{11} = A_{11} * B_{11}$

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$\begin{bmatrix} 9 & 4 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 7 & 6 \\ 3 & 9 \end{bmatrix}$$

$$P = (9+8)(7+9)$$

$$= (17)(16)$$

$$\boxed{P = 272}$$

$$Q = B_{11}(A_{21} + A_{22})$$

$$= 7(7+8)$$

$$= 7(15)$$

$$\boxed{Q = 105}$$

$$R = A_{11}(B_{12} - B_{22})$$

$$= 9(6-9)$$

$$= 9(-3)$$

$$\boxed{R = -27}$$

$$S = A_{22}(B_{21} - B_{11})$$

$$= 8(3-7)$$

$$= 8(-4)$$

$$\boxed{S = -32}$$

$$T = B_{22}(A_{11} + A_{12})$$

$$= 9(9+4)$$

$$= 9(13)$$

$$\boxed{T = 117}$$

$$U = (A_{21} - A_{11})(B_{11} + B_{12})$$

$$= (7-9)(7+6)$$

$$= (-2)(13)$$

$$\boxed{U = -26}$$

$$V = (B_{21} + B_{22})(A_{12} - A_{22})$$

$$= (3+9)(4-8)$$

$$= (12)(-4)$$

$$\boxed{V = -48}$$

(7)

$$C_{11} = P + S - T + V$$

$$C_{12} = R + T$$

$$C_{21} = Q + S$$

$$C_{22} = P + R - Q + U$$

$$\begin{array}{l|l} C_{11} = 272 + (-32) - 117 + (-48) & C_{12} = -27 + 117 \\ & \boxed{C_{12} = 90} \\ \boxed{C_{11} = 75} & \end{array}$$

$$\begin{array}{l|l} C_{21} = 105 + (-32) & C_{22} = 272 - 27 - 105 - 26 \\ & C_{22} = 114 \\ & \end{array}$$

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 75 & 90 \\ 73 & 114 \end{bmatrix}$$

$$C_{11} = \begin{bmatrix} 75 & 90 \\ 73 & 114 \end{bmatrix}$$

||y comput for A_{12} & B_{12} , A_{21} & B_{21} , A_{22} & B_{22}

$$A_{12} = \begin{bmatrix} 6 & 7 \\ 1 & 4 \end{bmatrix} \quad B_{12} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

$$\text{find } C_{12} = \begin{bmatrix} C_{13} & C_{14} \\ C_{23} & C_{24} \end{bmatrix}$$

$$\begin{aligned}
 P &= (A_{11} + A_{22})(B_{11} + B_{22}) \\
 &= (6 + 4)(2 + 3) \\
 &= (10)(5) \\
 \boxed{P} &= \boxed{50}
 \end{aligned}$$

$$\begin{aligned}
 R &= A_{11}(B_{12} - B_{22}) \\
 &= 6(1 - 3) \\
 &= 6(-2) \\
 \boxed{R} &= \boxed{-12}
 \end{aligned}$$

$$\begin{aligned}
 T &= B_{22}(A_{11} + A_{12}) \\
 &= 3(6 + 7) \\
 &= 3(13) \\
 \boxed{T} &= \boxed{39}
 \end{aligned}$$

$$\begin{aligned}
 V &= (B_{21} + B_{22})(A_{12} - A_{22}) \\
 &= (0 + 3)(7 - 4) \\
 &= 3(3) \\
 \boxed{V} &= \boxed{9}
 \end{aligned}$$

$$C_{13} = P + S - T + V$$

$$C_{14} = R + T$$

$$C_{23} = Q + S$$

$$C_{24} = P + R - Q + U$$

$$\begin{aligned}
 Q &= B_{11}(A_{21} + A_{22}) \\
 &= 2(1 + 4) \\
 &= 2(5) \\
 \boxed{Q} &= \boxed{10}
 \end{aligned}$$

$$\begin{aligned}
 S &= A_{22}(B_{21} - B_{11}) \\
 &= 4(0 - 2) \\
 \boxed{S} &= \boxed{-8}
 \end{aligned}$$

$$\begin{aligned}
 U &= (A_{21} - A_{11})(B_{11} + B_{12}) \\
 &= (1 - 6)(2 + 1) \\
 &= (-5)(3) \\
 \boxed{U} &= \boxed{-15}
 \end{aligned}$$

$$C_{13} = 50 + (-8) - 39 + 9 = 12$$

$$C_{14} = -12 + 39 = 27$$

$$C_{23} = 10 + (-8) = 2$$

$$C_{24} = 50 + (-12) - 10 - 15 = 13$$

$$\therefore \begin{bmatrix} C_{13} & C_{14} \\ C_{23} & C_{24} \end{bmatrix} = \begin{bmatrix} 12 & 27 \\ 2 & 13 \end{bmatrix} \quad C_{12} = \begin{bmatrix} 12 & 27 \\ 2 & 13 \end{bmatrix}$$

lly do for another matrix

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$$A_{21} = \begin{bmatrix} 4 & 3 \\ 5 & 3 \end{bmatrix} \quad B_{21} = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \quad \text{and } C_{21} = \begin{bmatrix} c_{31} & c_{32} \\ c_{41} & c_{42} \end{bmatrix}$$

$$P = (4+3)(2+2) = (7)(4) = 28$$

$$Q = 2(5+3) = (8)(2) = 16$$

$$R = 4(5+2) = (4)(3) = 12$$

$$S = 3(3-2) = 3(1) = 3$$

$$T = 2(4+3) = 14$$

$$U = (5-4)(2+5) = 7$$

$$V = (3-3)(3+2) = 0$$

$$c_{31} = 17$$

$$c_{32} = 26$$

$$c_{41} = 16+3=19$$

$$c_{42} = 31$$

$$\therefore \begin{bmatrix} c_{31} & c_{32} \\ c_{41} & c_{42} \end{bmatrix} = \begin{bmatrix} 17 & 26 \\ 19 & 31 \end{bmatrix}$$

$$C_{21} = \begin{bmatrix} 17 & 26 \\ 19 & 31 \end{bmatrix}$$

lly do it for another matrix

$$A_{22} = \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix} \quad B_{22} = \begin{bmatrix} 2 & 9 \\ 4 & 7 \end{bmatrix} \quad C_{22} = \begin{bmatrix} c_{33} & c_{34} \\ c_{43} & c_{44} \end{bmatrix}$$

$$P = (2+2)(2+7) = (4)(9) = 36$$

$$Q = (0+2) \cdot 2 = 4$$

$$R = 2(9-7) = 2(2) = 4$$

$$S = 2(4-2) = 2(2) = 4$$

$$T = (2+6)7 = (8)7 = 56$$

$$U = (0-2)(2+9) \\ = (-2)(11) = -22$$

$$V = (6-2)(4+7)$$

$$= 4(11)$$

$$= 44$$

$$\therefore C_{33} = 28$$

$$C_{34} = 60$$

$$C_{43} = 8$$

$$C_{44} = 14$$

$$\therefore \begin{bmatrix} C_{33} & C_{34} \\ C_{43} & C_{44} \end{bmatrix} = \begin{bmatrix} 28 & 60 \\ 8 & 14 \end{bmatrix}$$

Thus the product of matrices

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix} = \begin{bmatrix} 9 & 4 & 6 & 7 \\ 7 & 8 & 1 & 4 \\ 4 & 3 & 2 & 6 \\ 5 & 3 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 7 & 6 & 2 & 1 \\ 3 & 9 & 0 & 3 \\ 2 & 5 & 2 & 9 \\ 3 & 2 & 4 & 7 \end{bmatrix}$$

$$\therefore \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix} = \begin{bmatrix} 75 & 90 & 12 & 27 \\ 73 & 114 & 2 & 13 \\ 17 & 26 & 28 & 60 \\ 19 & 31 & 8 & 14 \end{bmatrix}$$